

Assignment for QA Automation Lead Software Engineer

Overview

This assignment aims to demonstrate your test planning, automation framework design, implementation, and people management skills while you work with our e-learning platform. We estimate it will take 6–10 hours to complete. If time is tight, prioritize the Test Plan and 2–3 automated tests. If you have any questions, email mike@learnworlds.com or constantinos@learnworlds.com.

Context

You can find documentation in [our support articles](#).
First, you need to [create a trial school](#) on our platform.

It is **essential** to create your school with:

- **email:** <yourFirstName>.<yourLastName>@learnworlds.com
- **Subdomain:** any available subdomain of your preference

Please do not forget your password. You will need it to access the system throughout this assignment.

Technologies required: Playwright, TypeScript, Page Object Model (POM) pattern.

Technical assessment

Based on LearnWorlds documentation and your trial school, you will need to:

Task 1: Test Plan

Create a **complete Test Plan** (Markdown or PDF) covering sandbox purchases of courses with and without the cart flow. You might also consider purchases with coupons in your scenarios.

Task 2: Test Data

Describe how you would seed the stage environment to run the tests. For the assessment, you might manually add data.

Task 3: Playwright Project

Select **2–3 test cases** from your plan and implement them in a Playwright project using **TypeScript**. Also, use the **Page Object Model (POM)**, **fixtures**, and the **latest Playwright methods** as best practices.

Task 3.1: Assertions

For any purchase test case you implement, you need to cover parts of the user details page (as admin navigate to <https://<<yourschool>>/author/users?tab=user> and select a user) and the payments page (<https://<<yourschool>>/author/sales>). This will help you assert specific details in your tests.

- User details page - Overview tab: user registration details and the number of courses enrolled in.
- User details page - Products tab: courses that the user is enrolled in, including some statuses.
- Payments page - Transactions tab: Purchase info (payment method used, date of purchase, etc.)

Task 3.2: API testing

Create an isolated test where you create a simple user by using an API call to the backend system. As a logged-in admin, you may use the endpoint **POST** https://<<yourschool>>/api/author/create_user.

- Implement the endpoint request payload and response schemas using TypeScript.
- Create a user object based on the request payload schema.
- Expect that the request was handled successfully.

Task 4: Reporting

Implement a reporting solution that shows a summary and per-test results. If this is not feasible or there is no time, Playwright HTML Reporter output is acceptable.

Task 5: CI Setup (Nice to Have)

Optional: Create a simple pipeline job (e.g. GitHub Actions, GitLab CI etc.) that runs the tests and creates a report. You may add the report to the job's artifacts or upload it to a host/reporter.

Team Lead skills

Please write short answers to the following questions.

Question 1

Tell us about a specific time when you mentored a junior engineer (or a peer) who was struggling with the quality or maintainability of their test code.

- How did you identify the issue?
- What specific techniques or tools did you introduce to help them level up?
- Looking back today, what would you have handled differently in your mentoring approach to make it more effective?

Question 2

Describe a real-life situation where you had **limited time** for coaching and code reviews while managing a team with **mixed experience levels**.

- How did you decide where to invest your time to ensure the whole team improved without compromising delivery?
 - What specific **criteria** did you use to prioritize your efforts, and why?
 - Looking back today, what is one thing you would have handled differently to better balance team growth with project deadlines?
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Hints

- [How to Make a Test Purchase \(Sandbox Mode\)](#)
 - [How to set up and use the cart feature in LearnWorlds](#)
 - AI can be used on any task; however, the final review and implementation should be yours.
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Final Submission

Please send us:

- Your school's URL
 - A **GitHub/GitLab repository** (preferred) or zipped folder containing:
 - Test Plan document
 - Test Data strategy (Task 2)
 - Playwright project source code
 - Reporting output or trace artifacts (if available)
 - **README.md** with setup, run instructions, and environment variables
 - CI config (if completed)
 - A document with your team lead skills Q&A
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Evaluation & Next Steps

We will review the completeness, clarity, and quality of your Test Plan, test data strategy, code structure (POM, fixtures), test stability, reporting, and your Team Lead skills responses. Candidates who meet our criteria will be invited to present and discuss their work in a follow-up call.

We wish you good luck and thank you for taking the time to work on this assignment 😊

A few words about LearnWorlds

Learning is the only human superpower. It's a gateway through which knowledge becomes a virtue, a value, a product, a skill, a prerequisite, a motive, and a competitive advantage.

LearnWorlds' vision is to enable individuals, businesses, and communities to leverage the power of learning, improving lives and cultivating a brighter, more resilient future for everyone.

Our mission is to stand out as the ultimate adaptable e-learning solution, tailored to meet the diverse, advanced and intricate business models of contemporary organizations. We are committed to delivering unmatched flexibility in our offerings, ensuring that businesses, individuals, and communities can effortlessly embrace knowledge empowerment as a cornerstone of their growth and success.

LearnWorlds enables its users to create complete e-learning websites, author online courses, and provides the tools to market and sell them as both B2B and B2C products. We have done this through the creation of a cloud-based e-learning platform that can be tailored to users' specific needs. Much more than just selling online courses, we have created everything required for a holistic and complete online learning experience, including but not limited to: a social

community, stylish templates, modular page builders, an assessment engine, automated certificates, and advanced analytics, all aimed at empowering both users and students.

LearnWorlds' white-label e-learning solution focuses on learners' needs, offering a holistic "world of learning." Created by academics, the platform is centered around true learning ecology, where students can immerse themselves in interactive content and learn from and with each other, reaping the benefits of a classroom setting accessible from anywhere. Our clients can create and distribute self-paced learning courses, with the three main current groups being: individual instructors aiming to sell their courses to the general public, organizations that wish to educate their employees, and partners or academic institutions. The platform has been created with the user at its heart; it strives to enhance learner satisfaction and forge deep, lasting connections between learners and instructors.

Today, LearnWorlds has c.12,000 users in more than 150 countries with 40% of customers coming from the US.

Please note that this assignment is solely for the purpose of evaluating your skills, approach, and thought process during the interview process. The work you submit will not be used, implemented, or distributed in any way by LearnWorlds. We respect the time and effort you put into this task and assure you that it is intended only to help us understand your capabilities for this role.